

EC2010

Macroeconomics: Money and Monetary Policy

Summer 2025 Examination · Paper 1
Duration 1.5 hours · Section A: 4 of 6 short questions (50%) · Section B: 1 of 2 long questions (50%)

Module	EC2010 — Macroeconomics: Money and Monetary Policy
Session	Summer 2025
Structure	Section A: six short questions, answer any four, each carrying equal weight (50% of the paper between them). Section B: two long questions, answer one, worth the remaining 50%.
This scheme covers	All six Section A questions and both Section B questions in full, so you can check your working whichever combination you sat.

Own-figure rule. If you carry an early error into a later part of a question but apply the right method consistently from that point on, you still pick up the marks for the later steps done correctly on your own figures.

A note on Section A weighting. Each Section A question is marked out of the raw total shown on the paper (e.g. A1 out of 100, split 50+50), then the four questions you select are combined and scaled so that Section A contributes 50% of your overall grade. The raw mark breakdowns below show you exactly how each question's own total is built up before that scaling is applied.

Section A · Question 1

SME Lending, Repayments & the Money Supply · 2 × 50 marks

A1100 marks

Explain how each of the following would be expected to affect the aggregate money supply in Ireland.

(i) Gross new lending advanced to SMEs by Irish Banks was €1.1 billion during Q3 2024.

(ii) Repayments by all SMEs to Irish Banks were €1.2 billion in Q3 2024.

In your answer you should show how each affects the assets and liabilities of a typical commercial bank.

Source: Central Bank of Ireland Statistical Release on Trends in SME and Large Enterprise Credit and Deposits: Q3 2024.

FULL MODEL ANSWER

Both parts turn on the same mechanism: when a commercial bank makes a new loan, it doesn't hand over pre-existing money — it creates a new deposit for the borrower at the moment the loan is drawn down. That new deposit is itself part of the money supply. Repayment runs the same mechanism in reverse and destroys money.

(i) New lending of €1.1bn

The bank grants the loan and credits the SME's current account with the funds. Nothing is transferred out of the bank's existing deposit base — a brand-new asset and a brand-new liability are created simultaneously.

Commercial bank balance sheet	Assets	Liabilities
Loans to SMEs	+€1.1bn	—
SME current account deposits	—	+€1.1bn
Net effect	+€1.1bn	+€1.1bn

The SME's current account deposit is part of M1/M3 (broad money). Since this deposit did not exist before the loan was made, the aggregate money supply **increases by €1.1bn**.

(ii) Repayments of €1.2bn

When SMEs repay, their current account is debited and the corresponding loan asset is written down by the same amount. The deposit that funded the repayment is extinguished, not transferred to the bank as retained cash.

Commercial bank balance sheet	Assets	Liabilities
Loans to SMEs	−€1.2bn	—
SME current account deposits	—	−€1.2bn
Net effect	−€1.2bn	−€1.2bn

This destroys €1.2bn of deposits, so the aggregate money supply **falls by €1.2bn**.

Worth noting: repayments (€1.2bn) exceeded new lending (€1.1bn) in the quarter, so the net effect of these two flows alone was a €0.1bn (€100m) contraction in the stock of money attributable to the SME lending channel — even though gross new lending was still positive. Don't confuse gross new lending with the net change in credit outstanding.

Mark Allocation — Part (i): New Lending (50 marks)

Tier	Marks	What separates this tier
40-50	Correct direction (increase) with a fully correct, correctly-labelled T-account/balance sheet entry showing both the asset (loan) and liability (deposit) side, and a clear statement that a new deposit is money.	
25-39	Correct direction and mostly correct balance sheet entries, but the explanation of why the deposit counts as new money is thin or the entries are only partially labelled.	
10-24	States the loan increases the money supply with little or no supporting balance sheet mechanics.	
0-9	Incorrect direction, or answer confuses this with a simple transfer of existing funds.	

Mark Allocation — Part (ii): Repayments (50 marks)

Tier	Marks	What separates this tier
40-50	Correct direction (decrease) with fully correct balance sheet entries on both sides and a clear statement that repayment destroys deposit money.	
25-39	Correct direction, entries mostly correct but explanation of the money-destruction mechanism underdeveloped.	
10-24	States money supply falls with little supporting mechanics.	
0-9	Incorrect direction or no meaningful balance sheet treatment.	

Question A1 Total: 100 Marks

Section A · Question 2

ECB Policy Rates & Communication · 30 + 70 marks

A2(a)**30 marks**

The following relates to the ECB's Monetary Policy Decisions, Press Release, Jan. 30th 2025: "The Governing Council today decided to lower the three key ECB interest rates by 25 basis points. Accordingly, the interest rates on the deposit facility, the main refinancing operations and the marginal lending facility will be decreased to 2.75%, 2.90% and 3.15% respectively." Explain the deposit facility.

A2(b)**70 marks**

The following is taken from the ECB's Monetary Policy Statement, Press Conference, Jan. 30th 2025: "We will continue to follow a data-dependent approach to determining the appropriate level and duration of restriction. In particular, our interest rate decisions will be based on our assessment of the inflation outlook in light of the incoming economic and financial data, the dynamics of underlying inflation and the strength of monetary policy transmission." Explain how this statement differs from Forward Guidance.

FULL MODEL ANSWER**(a) The deposit facility — 30 marks**

What it is: One of the ECB's two standing facilities (alongside the marginal lending facility). Eligible commercial banks can deposit surplus overnight liquidity directly with their national central bank on their own initiative, without needing a counterparty.

What it does: It sets the floor of the interest rate corridor. No bank will lend to another bank in the overnight market at a rate below what it can earn risk-free from its own central bank, so the deposit rate anchors the bottom of short-term money market rates.

Why it matters today: With the Eurosystem's balance sheet still holding a large stock of reserves left over from APP/PEPP, the banking system as a whole is long liquidity. In this environment (a de-facto floor system) the deposit facility rate, rather than the MRO rate, is effectively the ECB's operative policy rate — it is the rate that market rates such as €STR track most closely.

Rate (Jan 2025)	Level	Role
Marginal lending facility	3.15%	Ceiling — emergency overnight borrowing from the Eurosystem
Main refinancing operations (MRO)	2.90%	Reference/signalling rate on regular liquidity operations
Deposit facility	2.75%	Floor — rate earned on surplus reserves parked overnight

(b) "Data-dependent, meeting-by-meeting" vs. Forward Guidance — 70 marks

Forward guidance is a commitment about the future path of policy, made explicit in advance — either time-contingent (“rates will stay at their present level until at least [date]”) or state-contingent (“rates will stay low until inflation reaches X% and is judged durable”). Its entire purpose is to shape expectations of future rates today, so that longer-term market rates move even without a current change in the policy rate — it deliberately reduces uncertainty about the future reaction function by pre-committing to it.

The January 2025 statement explicitly does the opposite: it says decisions will be taken meeting-by-meeting based on whatever data arrives, with no pre-set path and no commitment to a particular level or duration of restriction beyond what the incoming data justifies. This preserves full optionality for the Governing Council rather than tying its hands.

Why the distinction matters forward guidance works by lowering long-run interest rate expectations (and hence long-term yields, mortgage rates, etc.) even before the ECB actually moves — it is itself a monetary policy instrument. A purely data-dependent, non-committal statement carries no such expectational stimulus: markets must continue to guess the path from data releases rather than from a stated commitment, so the statement is a description of the decision process, not a signal about the future path.

Common confusion to avoid: data-dependence and forward guidance are not opposites in the sense of one being ‘no information’ — the ECB is still telling markets *what it is watching* (underlying inflation dynamics, transmission strength). The key difference is that it is not telling markets *what it will do* in response, which is exactly what forward guidance does.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
24-30	Correctly identifies the deposit facility as a standing facility setting the floor of the corridor, with reference to the current context (excess liquidity / de-facto floor system).	
14-23	Correct basic definition but no discussion of its role as the effective floor or its relevance under current liquidity conditions.	
0-13	Vague or partially incorrect description (e.g. confused with MRO or marginal lending facility).	

Mark Allocation — Part (b) (70 marks)

Tier	Marks	What separates this tier
55-70	Clear, correct contrast: forward guidance is a pre-committed signal about the future path (time- or state-contingent) used to shape expectations; the January statement is explicitly non-committal and meeting-by-meeting. Explains why this distinction matters for expectations/transmission.	
35-54	Correctly identifies that the statement avoids commitment but the explanation of what forward guidance actually is/does is incomplete.	
15-34	General sense that the statement is 'cautious' or 'flexible' without a real definition of forward guidance to contrast it against.	
0-14	No meaningful distinction drawn.	

Question A2 Total: 100 Marks

Section A · Question 3

The Eurosystem Operational Framework · 100 marks

A3

100 marks

The following excerpt is taken from the Central Bank of Ireland signed article *The evolution of the Eurosystem operational framework and how recent changes may impact banks in Ireland* (2024): “This Article serves to explain what an operational framework is and why it is important for the smooth transmission of monetary policy. It describes the evolution of the Eurosystem operational framework, from inception as a ‘classic’ corridor system with scarce reserves to a de-facto floor system.” Analyse the Eurosystem transition from a corridor system to a de-facto floor system.

FULL MODEL ANSWER

An **operational framework** is the set of instruments (standing facilities, open market operations, reserve requirements) a central bank uses to steer short-term market interest rates towards its policy rate. Its effectiveness depends critically on how scarce or abundant reserves are in the banking system.

Interest Rate Corridor vs. De-Facto Floor System

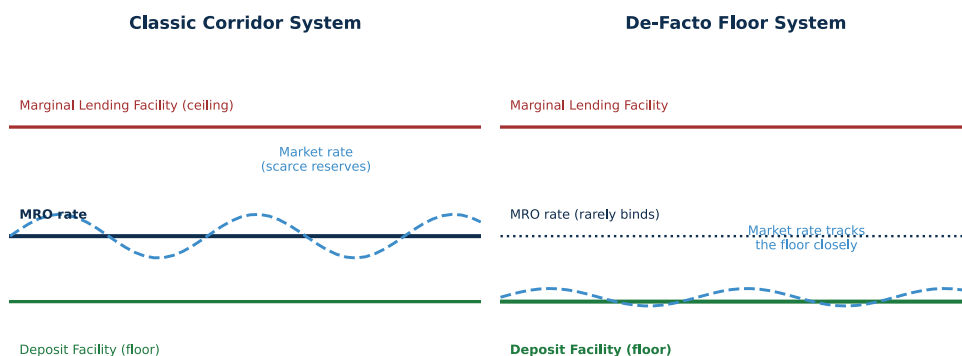


Fig. 1 — The interest rate corridor before and after the shift to abundant reserves

The classic corridor system (pre-2008)

Scarce reserves: the Eurosystem supplied just enough reserves through weekly MROs (variable-rate tender, banks bidding for liquidity) to meet the aggregate reserve requirement, deliberately keeping the system close to balance.

Active interbank market: because reserves were scarce, banks with a shortfall borrowed from banks with a surplus in the overnight market (EONIA), rather than routinely using the standing facilities.

Narrow corridor, market rate near the middle: the marginal lending facility (ceiling) and deposit facility (floor) were symmetric around the MRO rate and existed mainly as a backstop for occasional imbalances. The overnight market rate (EONIA) traded close to the MRO rate, in the middle of the corridor, because genuine scarcity forced continuous interbank trading and kept the market rate anchored near the policy rate.

The shift to a de-facto floor system (post-crisis)

Fixed-rate full allotment: from October 2008 the ECB moved to fixed-rate tenders with full allotment — banks could borrow as much as they wanted against eligible collateral at the fixed MRO rate, removing the deliberate scarcity that had underpinned the corridor.

Large-scale asset purchases: APP and PEPP injected a very large stock of reserves into the banking system directly (the Eurosystem buys bonds, crediting the seller's bank with new reserves). This left the system chronically *long* liquidity rather than roughly balanced.

Reserves now abundant, not scarce: with reserves far in excess of requirements, banks no longer need to borrow from each other overnight to square their positions — a bank with surplus reserves has nowhere better to place them than the deposit facility, since the interbank market offers little more than the deposit rate itself once reserves are abundant.

Market rate settles at the floor: consequently the overnight market rate (now €STR) trades very close to the deposit facility rate, not the MRO rate. The deposit rate, not the MRO rate, has effectively become the operative policy rate — hence 'de-facto floor system.'

Why the transition matters

Control of the operative rate: the ECB now signals and manages policy primarily via the deposit facility rate, and the January 2025 rate decision (Q. A2) is quoted precisely in terms of that rate.

Balance sheet size becomes a policy tool: as the Eurosystem's asset holdings (and hence excess reserves) shrink through QT, the system could eventually migrate back towards genuine scarcity, at which point the corridor's width and the MRO rate would matter again for where the market rate settles — the Eurosystem's 2024 operational framework review addresses exactly this transition path.

Reduced signalling role for the corridor width: because market rates hug the floor regardless of the corridor's width once reserves are abundant, the symmetry/asymmetry of the corridor (see A2(a): 15bp to the floor, 25bp to the ceiling) matters far less for short-term rate control than it did under scarcity.

Exam tip: a strong answer explicitly links *cause* (fixed-rate full allotment + APP/PEPP → abundant reserves) to *effect* (banks no longer need the interbank market → market rate gravitates to the deposit floor) rather than just describing both systems in isolation.

Mark Allocation (100 marks)

Tier	Marks	What separates this tier
75-100	Fully analyses both systems and the mechanism linking them: scarce reserves & active interbank trading under the corridor vs. abundant reserves (from full allotment / APP-PEPP) pushing the market rate to the deposit floor. Explains why this makes the deposit rate the effective policy rate today, with clear economic reasoning throughout.	
50-74	Describes both systems reasonably well but the causal link (why abundant reserves push the rate to the floor specifically, rather than just 'somewhere lower') is underdeveloped.	
25-49	Descriptive only — states that the system moved from a corridor to a floor without explaining the reserve-scarcity mechanism driving it.	
0-24	Minimal or largely incorrect description of either system.	

Question A3 Total: 100 Marks

Section A · Question 4

A Fall in Government Expenditure: The IS-TR Model · 100 marks

A4

100 marks

Use the IS-TR model to examine how a decrease in government expenditure could affect interest rates and economic activity in the short run. You should use a diagram to support your answer.

FULL MODEL ANSWER

Setting up the model

The **IS curve** plots combinations of output (Y) and the real interest rate (r) at which planned spending equals output: $Y = C(Y-T) + I(r) + G + NX$. It slopes downward because a higher real interest rate depresses interest-sensitive spending (mainly investment), reducing equilibrium output.

The **TR curve** (the central bank's reaction/Taylor-rule function — see Q. A5) slopes upward: the central bank raises the real interest rate as output rises above trend (to contain the resulting inflationary pressure) and cuts it as output falls below trend.

Step 1 — The direct effect of the fall in G

Government expenditure (G) is a direct, autonomous component of planned spending. A cut in G reduces planned spending at every level of the real interest rate, so the entire IS curve shifts left (from IS to IS'). The TR curve is unaffected — the central bank's reaction function itself hasn't changed, only the position of aggregate demand.

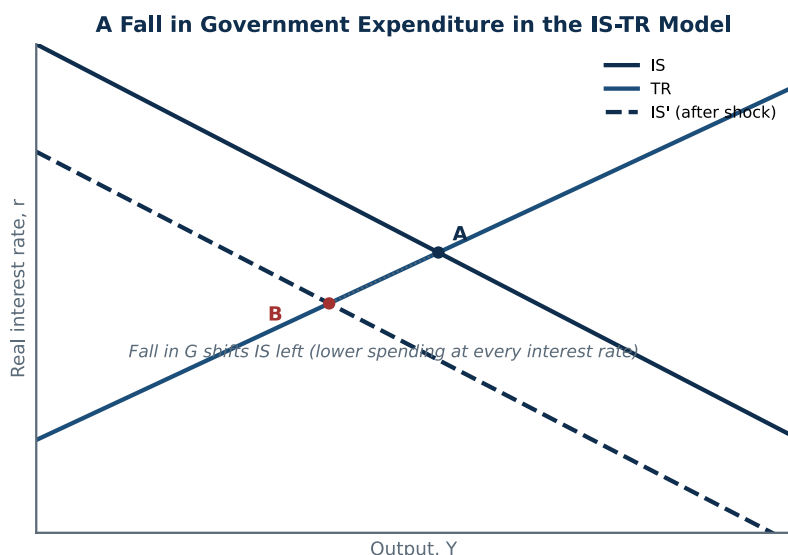


Fig. 1 — A fall in G shifts IS left; the economy moves from A to B along the (unchanged) TR curve

Step 2 — The new short-run equilibrium

The economy moves from the initial equilibrium (A) to a new equilibrium (B), which lies on the original TR curve but on the new, further-left IS' curve. Two things happen simultaneously:

- **Output falls.** At the old real interest rate, planned spending is now below output, so firms cut back production — output contracts (the standard fiscal multiplier effect, before any monetary offset).
- **The real interest rate falls.** Moving along the TR curve, lower output means

the central bank is now facing weaker demand/inflationary pressure than before, so it responds by cutting the real interest rate — this is not the central bank fighting the shock, it is the central bank's *systematic* response to lower output and lower inflationary pressure, as described by the reaction function itself.

Net effect: output falls, but by less than it would under a fixed interest rate, because the endogenous fall in r (moving down the TR curve) partially offsets the initial contraction by stimulating some interest-sensitive spending (investment) back. The TR curve therefore acts as an automatic stabiliser relative to a world with a constant interest rate.

Diagram guidance: one diagram is sufficient — a single IS-TR panel showing the IS curve shifting left with the equilibrium sliding down the unchanged TR curve is all that's required; you don't need a second AD/AS panel unless the question explicitly asks for both models (compare Q. B2(a)).

Mark Allocation (100 marks)

Tier	Marks	What separates this tier
75-100	Correct diagram (IS shifts left, TR unchanged, new equilibrium on the original TR curve) with a full explanation of both the fall in output and the fall in the real interest rate as a movement along TR, and why the interest-rate response partially cushions the output fall.	
50-74	Correct diagram and correctly identifies that both Y and r fall, but the explanation of <i>why</i> r falls (i.e. that it is the central bank's systematic response along TR, not an independent shock) is weak or missing.	
25-49	Identifies that output falls but the model is not properly set up (e.g. TR curve omitted, or IS/TR roles confused) and/or no diagram.	
0-24	Minimal or largely incorrect treatment.	

Question A4 Total: 100 Marks

Section A · Question 5

Interpreting a Central Bank Reaction Function · 100 marks

A5**100 marks**

The following equation represents the reaction function of a Central Bank:

$$i = i^n + 1.5(\pi - \pi^T) + 0.5(Y - Y^T)/Y^T$$

where i^n = Natural (Neutral) Rate of Interest, π^T = Inflation Target, Y^T = Trend Output. Explain how the Central Bank changes interest rates, according to this reaction function.

FULL MODEL ANSWER**Reading the equation**

The equation says the central bank sets its policy rate (i) equal to a baseline — the neutral rate (i^n), the rate consistent with output at trend and inflation at target — plus two adjustment terms that respond to how far the economy currently sits from that benchmark.

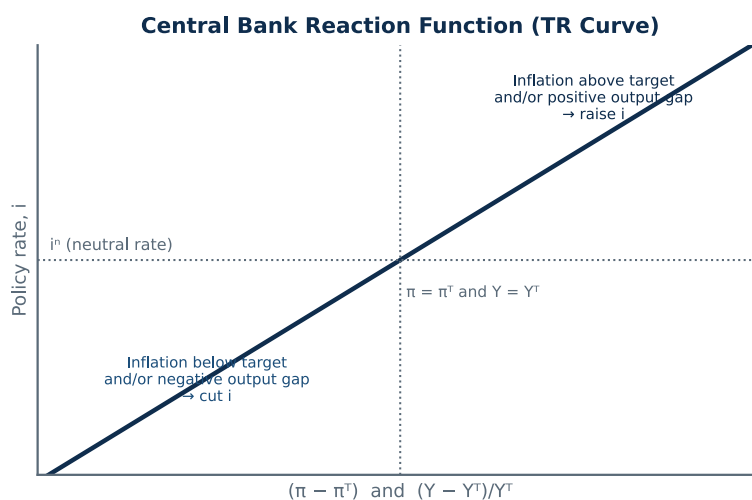


Fig. 1 — The TR curve implied by this reaction function

The two gap terms

Inflation gap, $1.5(\pi - \pi^T)$: if actual inflation is above target, this term is positive and the bank raises the policy rate above neutral. The coefficient of **1.5** means the rate rises by more than one-for-one with the inflation gap — a 1 percentage point rise in inflation above target triggers a 1.5pp rise in the nominal policy rate. Because the real rate is (approximately) $i - \pi$, a more-than-one-for-one nominal response is required to actually raise the real rate and cool demand — a coefficient below 1 would let the real rate fall as inflation rises, which would be destabilising. This is the **Taylor principle**.

Output gap, $0.5(Y - Y^T)/Y^T$: if output is above trend ($Y > Y^T$), the term is positive and the bank also raises rates, anticipating the inflationary pressure that a positive output gap tends to generate; if output is below trend, this term is negative and the bank cuts rates to support demand. The coefficient of **0.5** is smaller than the inflation coefficient, so, taken in isolation, the central bank responds less aggressively (in percentage points) to a given percentage output gap than to an equivalent percentage-point inflation gap.

Putting it together

- When $\pi = \pi^T$ and $Y = Y^T$ (the economy is exactly at target on both dimensions), both bracket terms are zero and $i = i^n$ — the bank simply holds the neutral rate.
- Any deviation of inflation from target, or of output from trend, moves the policy rate away from neutral in the direction that pushes the economy back towards target — the reaction function is inherently stabilising, which is exactly what generates the upward-sloping TR curve used throughout Topic 4 (Q. A4) and Topic 5 (Q. A6, B2).
- If both gaps are positive at once (an overheating economy with above-target inflation), the two terms reinforce each other and the rate rise is larger than either gap would produce alone.

Illustrative check: with $i^n = 2\%$, $\pi^T = 2\%$: inflation 1pp above target on its own adds 1.5pp to i ($i = 3.5\%$); output 2% above trend on its own adds 1.0pp to i ($i = 3.0\%$); both together add 2.5pp ($i = 4.5\%$). These numbers aren't required for full marks — what matters is correctly explaining the direction and relative size of each term's effect.

Mark Allocation (100 marks)

Tier	Marks	What separates this tier
75-100	Correctly explains both gap terms (direction and relative magnitude of the 1.5 vs 0.5 coefficients), explicitly identifies the Taylor principle (coefficient on inflation >1 needed to raise the real rate), and states the baseline case ($i = i^n$ when both gaps are zero).	
50-74	Correctly explains that the bank raises/cuts rates in response to each gap but doesn't discuss why the inflation coefficient needs to exceed 1 (the Taylor principle), or omits the baseline case.	
25-49	General description of 'the bank raises rates when inflation/output is high' without engaging with the structure or coefficients of the equation.	
0-24	Minimal or largely incorrect interpretation of the equation.	

Question A5 Total: 100 Marks

Section A · Question 6

A Negative Energy Shock: The AD/AS Model · 100 marks

A6

100 marks

Use an appropriate model to examine how a negative energy shock (e.g. higher energy prices) could affect economic activity and prices over the medium run. You should use a diagram to support your answer.

FULL MODEL ANSWER

Setting up the model

The appropriate tool here is the **AD/AS model** in (output, inflation) space, built from Topic 5. The AD curve (downward sloping) is derived by combining the IS and TR relationships from Topic 4: it traces out the combinations of output and inflation consistent with goods-market equilibrium once the central bank's systematic interest rate response is accounted for. The AS curve (upward sloping in the short run) reflects firms' price-setting behaviour: at higher output, tighter labour/product markets push costs and prices up.

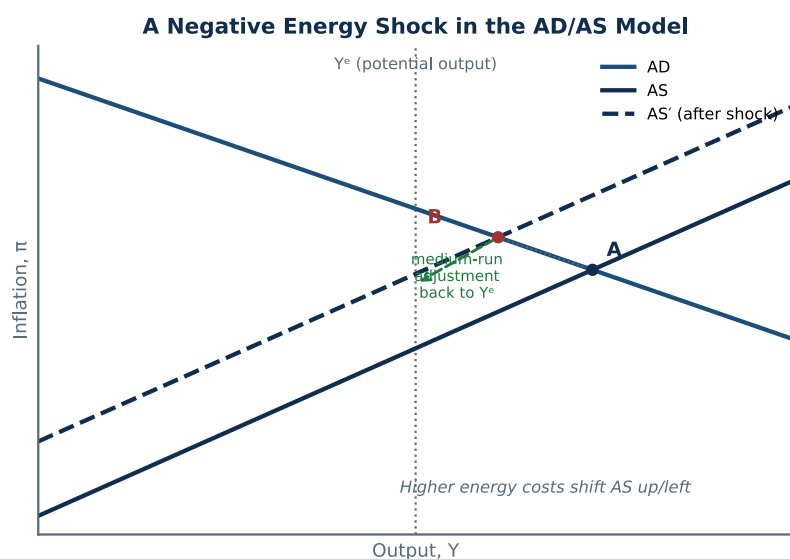


Fig. 1 — A negative energy shock shifts AS left; the economy moves from A to B, then adjusts back towards Y^e over the medium run

Step 1 — The short-run impact

A rise in energy prices is a direct increase in firms' input costs. At any given level of output, firms now need to charge a higher price to cover costs — this shifts the **AS curve up/left**. The AD curve is unaffected in the first instance (the shock originates on the supply side).

Result: the new short-run equilibrium (B) has higher inflation and lower output than the initial equilibrium (A) — the textbook **stagflation** outcome. This is the key feature that distinguishes a supply shock from a demand shock: a demand shock moves output and inflation in the same direction; a supply shock moves them in opposite directions.

Step 2 — The medium-run adjustment

Output at B is below potential output (Y^e), which over time puts downward pressure on wages and non-energy costs (via the Phillips-curve/expectations mechanism

underlying the AS curve), gradually shifting AS back towards its original position as expectations and cost pressures adjust — provided the energy price shock itself does not repeat and inflation expectations remain anchored. In the medium run, absent further shocks, output returns to Y^e and inflation returns towards target, but the price level is permanently higher than it would otherwise have been.

The role of monetary policy: if the central bank instead responds to the inflation spike by tightening (raising i along its TR curve), this shifts AD left as well, which brings inflation down faster but pushes output further below potential in the short run — the classic policy trade-off facing a central bank after a supply shock (compare this to a genuine demand-driven tightening in Q. B2(a), where AD and AS both move consistently in one direction rather than pulling against each other).

Second-round effects: if workers/firms build the higher energy-driven inflation into wage demands and pricing before it fades, the AS shift can become persistent rather than temporary — this is why central banks watch measures of underlying/core inflation and inflation expectations closely after an energy shock (see also Q. A2(a) on data-dependence and Q. B1(b) on the post-pandemic inflation episode).

Diagram guidance: a single AD/AS panel showing AS shifting left, the short-run equilibrium at B, and an arrow indicating the medium-run drift back towards Y^e is sufficient — you do not need a separate IS-TR panel unless a question explicitly requires both models together (see Q. B2(a)).

Mark Allocation (100 marks)

Tier	Marks	What separates this tier
75-100	Correct AD/AS diagram with AS shifting left, correctly identifies the short-run stagflation outcome (higher inflation, lower output), and explains the medium-run adjustment back towards potential output, including the policy trade-off if the central bank tightens.	
50-74	Correct diagram and identifies the short-run stagflation outcome, but the medium-run adjustment mechanism or the policy trade-off is underdeveloped or missing.	
25-49	Recognises that prices rise and/or output falls but the model/diagram is not correctly constructed (e.g. wrong curve shifted, or AD/AS confused with a simple demand-side story).	
0-24	Minimal or largely incorrect treatment.	

Question A6 Total: 100 Marks

Section B: Long Questions

Answer 1 of 2 · this scheme covers both in full

Section B questions are broader and typically integrate several ideas together. Both questions below are marked to the same standard as Section A — full model answers with tiered mark bands — so you can see exactly what a complete answer to either would look like.

Section B · Question 1(a)

How the Central Bank Influences Money Creation · 2 × 30 marks

B1(a)**60 marks**

The Central Bank can influence money creation directly and indirectly. Explain how the Central Bank can influence the money supply in the following ways:

- (i) Setting policy interest rates.
- (ii) Purchasing Government bonds from a pension fund.

In (ii), you should demonstrate your understanding by showing how bond purchases affect the assets and liabilities of both the Central Bank and the pension fund.

FULL MODEL ANSWER**(i) Setting policy interest rates — 30 marks**

Indirect channel: the policy rate doesn't itself create or destroy money directly — it works through the transmission mechanism. A change in the policy rate moves money-market rates (see Q. A2/A3), which commercial banks pass through to their own lending and deposit rates.

The credit channel: a lower policy rate lowers the cost of borrowing, raising the demand for loans; since new bank lending creates new deposits (Q. A1), more lending means more money created. A higher policy rate works in reverse, discouraging borrowing and slowing money creation.

The signalling/expectations channel: the policy rate (and how the Central Bank communicates around it — see forward guidance, Q. A2(b)) also shapes expectations of future rates, spending and inflation, which feeds back into current borrowing and spending decisions even before any loans are actually taken out.

(ii) Purchasing Government bonds from a pension fund — 30 marks

This is a **direct** channel of influence — an open market operation / asset purchase that mechanically expands both balance sheets involved. Say the Central Bank buys €500m of government bonds held by a pension fund. The pension fund does not hold an account directly at the Central Bank, so the transaction settles through the pension fund's commercial bank.

Central Bank balance sheet	Assets	Liabilities
Government bonds purchased	+€500m	—
Reserves credited to pension fund's commercial bank	—	+€500m
Net effect	+€500m	+€500m

Pension fund balance sheet	Before	After
Government bonds	€500m	€0m
Bank deposit	€0m	€500m
Total assets	€500m	€500m

Net result: the pension fund's total assets are unchanged — it has simply swapped a bond for a bank deposit. But the commercial bank now holds a new deposit liability (owed to the pension fund) matched by a new reserve asset (held at the Central Bank), and the Central Bank's own balance sheet has expanded on both sides. Both base money (reserves) and broad money (bank deposits) rise by €500m — new money has been created, even though no new lending to a household or firm was involved.

Contrast with (i): setting the policy rate works *indirectly*, by changing the incentives banks and borrowers face; a bond purchase works *directly*, by mechanically injecting new reserves and (via the seller's deposit) new broad money into the system the moment the transaction settles. Large-scale versions of exactly this transaction are what built up the reserve stock behind the shift to a de-facto floor system in Q. A3.

Mark Allocation — Part (i) (30 marks)

Tier	Marks	What separates this tier
24-30	Correctly identifies this as an indirect/transmission-mechanism channel and explains how the rate change affects borrowing costs and hence loan demand and money creation.	
14-23	States that a rate cut/rise increases/decreases money supply without explaining the credit-channel mechanism.	
0-13	Minimal or incorrect treatment.	

Mark Allocation — Part (ii) (30 marks)

Tier	Marks	What separates this tier
24-30	Fully correct balance sheet treatment for both the Central Bank and the pension fund (via the commercial bank), correctly showing that base money and broad money both rise by the purchase amount.	
14-23	Broadly correct direction of the balance sheet changes but the mechanics (e.g. the role of the commercial bank, or which side of the pension fund's balance sheet changes) are incomplete.	
0-13	States that the money supply rises without correct balance sheet entries.	

Section B · Question 1(b)

The Quantity Theory of Money and Recent Euro Area Inflation · 40 marks

B1(b)**40 marks**

“As such the recent experience serves as a reminder that the Quantity Theory of Money is not a vacuous concept of no practical importance for modern day central banks.” (Ref: Thünen Lecture by Isabel Schnabel, Member of the Executive Board of the ECB, Verein für Socialpolitik, Regensburg, 25th Sept. 2023.) Does the Quantity Theory of Money help us to understand the subdued price inflation performance in the 2015-2018 period in the Euro Area and the post-pandemic (2021, 2022) rise in price inflation in the Euro Area? Explain your answer. In your answer, you should use economic theory, your knowledge of how money is created and relevant readings in EC2010 to support your answer.

FULL MODEL ANSWER**The Quantity Theory of Money as the framework**

The Quantity Theory starts from the identity $MV = PY$ (money stock $\times$ velocity = price level $\times$ real output), which becomes a theory once velocity (V) is treated as roughly stable: on that assumption, growth in the money supply should translate fairly directly into growth in nominal spending, and — once real output growth is netted out — into inflation. The Regensburg lecture's argument is that the QTM was too readily dismissed during the low-inflation 2010s, and that the post-pandemic experience shows the underlying mechanism was never dead, only temporarily obscured by velocity behaviour.

2015-2018: subdued inflation despite large asset purchases

What happened to M: the ECB's Asset Purchase Programme (APP), launched in 2015, expanded the monetary base substantially — reserves rose sharply as the Eurosystem bought large quantities of government and corporate bonds (mechanically, exactly the balance sheet expansion shown in Q. B1(a)(ii), just at much larger scale).

But inflation stayed subdued: headline and underlying inflation remained persistently below the ECB's target for most of this period, which on a naive reading looks like a QTM failure — large M growth, weak P growth.

The QTM-consistent resolution: velocity fell. much of the new reserve money created did not flow into broad money and spending at the same pace; banks held a large share of the new reserves rather than expanding lending proportionally (weak credit demand after the sovereign debt crisis, deleveraging, and a weak money multiplier), and household/firm money holdings rose relative to spending (increased money demand, i.e. lower velocity) in a low-growth, low-confidence environment. A fall in V offsetting a rise in M is entirely consistent with the QTM identity — it is not evidence against the theory, but it does mean M growth alone is a poor *short-run* predictor of inflation when V is unstable.

2021-2022: the post-pandemic inflation surge

Money growth accelerated further and faster: pandemic-era fiscal support (financed in large part by PEPP purchases) pushed broad money growth (M3) to historically high rates, this time flowing more directly into household deposits (furlough/support schemes, forced saving during lockdowns) rather than sitting as idle bank reserves.

Velocity rebounded as economies reopened: pent-up demand and the release of accumulated savings once restrictions lifted meant V rose again, reinforcing rather than offsetting the effect of high M growth — the QTM mechanism reasserted itself once both M and V were moving in the same direction.

Supply-side shocks compounded the picture: the energy price shock (Q. A6) and pandemic-related supply chain disruption pushed AS left at the same time as strong monetary-driven demand pushed AD along a QTM-consistent path — so the 2021-2022 inflation surge reflects both a demand-side QTM story and a genuine supply shock, not one or the other in isolation.

Verdict

Partially, but not mechanically

The QTM is useful as an organising framework — it correctly points to money growth as a driver of inflation over a longer horizon, and the 2015-2018 episode is explained, not contradicted, once velocity instability is taken into account. But it is not a reliable short-run forecasting tool on its own: the same monetary expansion produced very different inflation outcomes in 2015-2018 versus 2021-2022 because velocity and the composition of money creation (reserves vs. broad money reaching households) differed sharply between the two episodes, and because supply shocks played an independent role in the second episode. This is precisely Schnabel's point: the QTM is not vacuous, but using it well requires taking velocity and the transmission of money creation into the real economy seriously, not just watching the level of M.

What a strong answer needs: explicit use of the $MV=PY$ identity, a clear account of why velocity moved differently in each period (not just an assertion that it did), and integration of the money-creation mechanics from earlier in the course (bank lending channel, QE mechanics) rather than treating QTM as a black box.

Mark Allocation (40 marks)

Tier	Marks	What separates this tier
30-40	Correctly sets out $MV=PY$, explains the 2015-2018 episode via a fall in velocity offsetting reserve growth (with reference to the credit/money-creation mechanism), explains the 2021-2022 episode via faster broad money growth reaching households plus a velocity rebound, discusses the compounding role of supply shocks, and reaches a clear, well-supported verdict.	
20-29	Uses the QTM framework and covers both episodes but the explanation of velocity's role, or the money-creation mechanics behind each episode, is thin.	
10-19	General discussion of inflation drivers in both periods without meaningfully engaging with the Quantity Theory or $MV=PY$.	
0-9	Minimal or off-topic answer.	

Question B1 Total: 100 Marks (60 + 40)

Section B · Question 2(a)

A Policy Rate Increase: IS-TR and AD/AS Together · 60 marks

B2(a)**60 marks**

Use the IS-TR model and the Aggregate Demand/Aggregate Supply model to show how an increase in the policy interest rate will have a contractionary effect on economic activity and reduce price inflation. You should use diagrams to support your answer.

FULL MODEL ANSWER

This question deliberately asks for both models together, because they answer two different halves of the same question: the IS-TR diagram shows *how a policy rate rise reduces output* in the short run; the AD/AS diagram shows *what that fall in output then does to inflation*. Used together they trace the full transmission chain from the policy decision to the inflation outcome.

Step 1 — IS-TR: the policy rate rise reduces output

An autonomous increase in the policy rate (over and above what the existing reaction function would already prescribe at current output/inflation) is modelled as an upward shift of the TR curve: at every level of output, the central bank is now setting a higher real interest rate than before. The IS curve itself is unaffected.

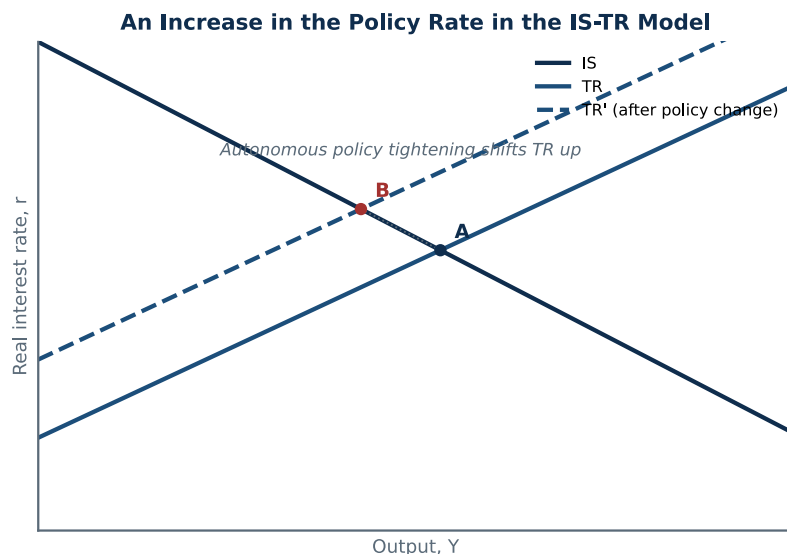


Fig. 1 — A tightening shifts TR up; the economy slides up the (unchanged) IS curve from A to B

Mechanism: the higher real interest rate raises the cost of borrowing and the return to saving, which discourages interest-sensitive spending — principally investment, but also durable consumption. Firms and households cut back, and the economy moves up the IS curve to a new equilibrium (B) with a higher real interest rate and lower output.

Step 2 — AD/AS: lower output feeds through to lower inflation

The AD curve in (output, inflation) space is itself built from the IS-TR relationship — it summarises, for each inflation rate, the level of output implied once the central bank's systematic rate response is taken into account. A policy tightening therefore shows up as the AD curve shifting left.

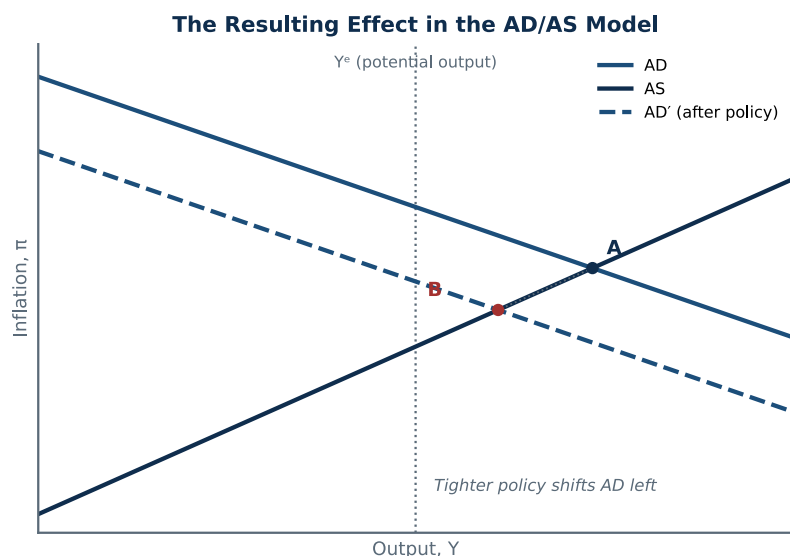


Fig. 2 — AD shifts left along the (unchanged) AS curve; the economy moves from A to B with lower output and lower inflation

Mechanism: with AD to the left and AS unchanged, the new equilibrium (B) lies at both lower output and lower inflation. This is the key contrast with the supply shock in Q. A6: here output and inflation move in the same direction (both fall), because the shock originates on the demand side, not the supply side.

Putting the two diagrams together

- The IS-TR diagram identifies the *source* of the shock (a deliberate policy tightening) and its *immediate* effect on real activity and the real interest rate.
- The AD/AS diagram then shows the *consequence* for inflation: lower output (from Step 1) reduces the pressure on wages and prices, so inflation falls along the AS curve.
- Together, the two diagrams tell the standard monetary transmission story: policy rate $\uparrow \rightarrow$ real interest rate $\uparrow \rightarrow$ interest-sensitive spending $\downarrow \rightarrow$ output $\downarrow \rightarrow$ inflation $\downarrow$.

Diagram guidance: this is one of the few EC2010 questions that explicitly requires two diagrams — most questions (Q. A4, Q. A6) need only one. Make sure both panels are drawn and clearly labelled A (before) and B (after), and that the direction of each shift is explained, not just shown.

Mark Allocation (60 marks)

Tier	Marks	What separates this tier
45-60	Both diagrams correctly drawn (TR shifts up in the IS-TR panel; AD shifts left in the AD/AS panel) with a clear explanation of the transmission chain linking the policy rate rise to lower output and then to lower inflation.	
30-44	Both diagrams present and broadly correct, but the link between the two (i.e. that the AD shift is the AD/AS-space consequence of the same IS-TR shock) is not made explicit.	
15-29	Only one of the two required diagrams/models is correctly used, or both are attempted but with significant errors in direction of shift.	
0-14	Minimal or largely incorrect treatment.	

Section B · Question 2(b)

Uncertainty in the ECB's Monetary Policy Process · 40 marks

B2(b)**40 marks**

Explain how uncertainty is incorporated into the monetary policy process at the European Central Bank (ECB).

FULL MODEL ANSWER

Central banks never observe the true state of the economy in real time — trend output, the neutral interest rate, the size of an output gap, and even current inflation dynamics are all estimated with error, and the effect of a rate change takes time to show up and is itself uncertain in size. The ECB builds several distinct responses to this uncertainty into its policy process.

Gradualism and caution (Brainard conservatism)

The idea: when a policymaker is uncertain about the size of the effect of its own instrument (multiplicative/parameter uncertainty, as opposed to just uncertainty about the state of the economy), the theoretically optimal response is to move more cautiously/gradually than it would under certainty, rather than to act aggressively on a point estimate that could be wrong.

In practice: the ECB typically moves in a sequence of measured steps (e.g. the 25bp increments referenced in Q. A2) rather than adjusting the whole way to its estimated optimal rate in one move, partly reflecting this logic.

Staff projections with explicit scenario/risk analysis

Quarterly macroeconomic projections: the ECB publishes baseline forecasts for growth and inflation, but explicitly surrounds them with alternative scenarios and risk assessments (upside/downside risks to the baseline) rather than presenting a single point forecast as certain.

Sensitivity analysis: projections are regularly stress-tested against alternative assumptions (e.g. energy prices, exchange rates) so the Governing Council can see how much the outlook would change under different states of the world.

A broad information set, not a single indicator

Two-pillar heritage: the ECB's strategy has historically combined economic analysis (short-to-medium-term growth/inflation dynamics) with monetary analysis (money and credit developments, directly relevant to the QTM discussion in Q. B1(b)), precisely so that policy isn't hostage to uncertainty or noise in any single indicator.

Underlying/core inflation measures: the January 2025 statement (Q. A2(b)) explicitly references 'the dynamics of underlying inflation' — looking through volatile components (energy, food) to reduce the risk of over- or under-reacting to noisy headline data.

Data-dependence and meeting-by-meeting decision-making

Direct link to Q. A2(b): the shift away from calendar-based or level-based forward guidance towards a purely data-dependent, meeting-by-meeting approach is itself a response to elevated uncertainty — after the post-pandemic/energy-shock period, pre-committing to a specific future path became riskier, so the ECB deliberately preserved optionality to adjust as new data arrives rather than being locked into a stated path that events might invalidate.

Communicating uncertainty itself

Fan charts and qualitative risk language: official communications describe the balance of risks around the baseline (e.g. 'risks to growth are tilted to the downside') rather than presenting forecasts as precise, which shapes market expectations around a distribution of outcomes rather than a single number.

What ties this together: every mechanism above is a structural response to *not knowing* the true state of the economy or the true effect of policy with precision — gradual moves guard against parameter uncertainty, scenario analysis guards against forecast uncertainty, a broad information set guards against indicator-specific noise, and data-dependence guards against being locked into a path overtaken by events.

Mark Allocation (40 marks)

Tier	Marks	What separates this tier
30-40	Identifies at least three distinct mechanisms (e.g. gradualism, scenario/risk analysis in projections, a broad information set, data-dependent meeting-by-meeting decisions) and explains <i>why</i> each is a rational response to a specific type of uncertainty, not just a list of ECB practices.	
20-29	Identifies two or more relevant mechanisms with reasonable explanation, but doesn't connect them back to the underlying idea of responding to uncertainty.	
10-19	Vague general discussion of ECB caution/communication without specific mechanisms.	
0-9	Minimal or off-topic answer.	

Question B2 Total: 100 Marks (60 + 40)